

Jayonto Dutta Plabon

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Research interests: Large Language Models for Low-Resource Languages, Federated Learning, Explainable AI, and Privacy-Preserving Machine Learning.

🎓 Education

2024–Present Expected: Jul'26	M.Sc. in Computer Science and Engineering (Running) University of Asia Pacific, Bangladesh Supervisor: Dr. Shah Murtaza Rashid Al Masud 🎓 Thesis: <i>BanglaLex: A Fine-Tuned Large Language Model for Bangladeshi Legal Advisory with Retrieval-Augmented Generation and Hallucination Grounding (Expected Completion: June 2026)</i>	CGPA: 4.00*/4.00
2020–2024	B.Sc. in Computer Science and Engineering University of Asia Pacific, Bangladesh Supervisor: Dr. Alope Kumar Saha 🎓 Thesis: <i>Explainable federated learning for privacy-preserving bangla sign language detection — achieved 98.36% classification accuracy using VGG19 with FedAvg across federated clients, eliminating reliance on centralized servers.</i>	CGPA: 3.87/4.00
2017–2019	Higher Secondary Certificate Examination Pangsha Govt. College, Bangladesh	GPA: 4.58/5.00
2015–2017	Secondary School Certificate Examination Yakub Ali Chowdhury Bidhapith, Bangladesh	GPA: 5.00/5.00

💼 Employment

Jul 25–Now	Lecturer Dept. of Computer Science and Engineering, University of Asia Pacific Teaching undergraduate courses including Computer Fundamentals (50+ students) and Competitive Programming (60+ students); mentoring 8+ capstone projects on federated learning and NLP applications.	Dhaka Bangladesh
Jul 24–Jan 25	Teaching Assistant Dept. of Computer Science and Engineering, University of Asia Pacific Served as TA for the OOP and DSA Lab; conducted weekly lab sessions for 60+ students, graded assignments, and provided one-on-one guidance during office hours.	Dhaka Bangladesh
Jan 24–May 24	Research Assistant Advance Machine Intelligence Research Lab Designed and implemented federated model training pipelines for emoji prediction tasks; contributed to experimental methodology and result analysis on published study (J2).	Dhaka Bangladesh

📖 Publication

Journal Articles

- 2025 J4 **From centralization to decentralization: Blockchain's role in transforming social media platforms (Review Article)**
BS Diba, JD Plabon, N Tasnim, M Hossain, D Mistry, S Sarker, MF Mridha, Y Okuyama, J Shin
IEEE Access (Q2, Impact Factor: 3.6) – 10 citations
DOI: [10.1109/ACCESS.2025.3561663](https://doi.org/10.1109/ACCESS.2025.3561663)

- J3 **Open problems and challenges in federated learning for IoT: A comprehensive review and strategic guide (Review Article)**
BS Diba, **JD Plabon**, TJ Mowla, N Nahar, D Mistry, S Sarker, MF Mridha, J Shin
Computers and Electrical Engineering (Q1, Impact Factor: 4.9) - 17 citations
DOI: [10.1016/j.compeleceng.2025.110515](https://doi.org/10.1016/j.compeleceng.2025.110515)
- 2024 J2 **Federated Learning-Based Architecture for Personalized Next Emoji Prediction for Social Media Comments**
D Mistry, **JD Plabon**, BS Diba, MSH Mukta, MF Mridha
IEEE Access (Q2, Impact Factor: 3.6) - 17 citations
DOI: [10.1109/ACCESS.2024.3448470](https://doi.org/10.1109/ACCESS.2024.3448470) ; Implementation: 
- J1 **Explainable federated learning for privacy-preserving bangla sign language detection**
BS Diba, **JD Plabon**, MDM Rahman, D Mistry, AK Saha, MF Mridha
Engineering Applications of Artificial Intelligence (Q1, Impact Factor: 8.0) - 20 citations
DOI: [10.1016/j.engappai.2024.108657](https://doi.org/10.1016/j.engappai.2024.108657) ; Implementation: 

Conference Proceedings

- 2025 C1 **Personalized Pomodoro Productivity Tracking with Privacy-Preserving Machine Learning in Human-Computer Interaction**
D Mistry, BS Diba, **JD Plabon**, SU Ahmed, MF Mridha
International Conference on Human-Computer Interaction
DOI: [10.1007/978-3-031-93965-5_29](https://doi.org/10.1007/978-3-031-93965-5_29)

Book Chapters

- 2025 Ch3 **Privacy-Preserving Kidney Stone Detection from X-Ray Image Using Federated Learning**
JD Plabon, MM Rahaman, MR Tonmoy, MG Kibriya, A Ishraq
Book: Federated Learning in Health Care Technology. **Publisher:** Springer Nature Singapore
DOI: [10.1007/978-981-96-8353-6_4](https://doi.org/10.1007/978-981-96-8353-6_4)
- Ch2 **Transforming Healthcare Analytics: Federated Learning for Privacy-Preserving Liver Lesion Detection and Classification**
MS Islam, **JD Plabon**, N Nahar, N Tasnim, SMRA Masud, J Shin, MR Tonmoy
Book: Federated Learning in Health Care Technology. **Publisher:** Springer Nature Singapore
DOI: [10.1007/978-981-96-8353-6_3](https://doi.org/10.1007/978-981-96-8353-6_3)
- Ch1 **Federated Learning Strategies for Confidential Leukemia Detection from Medical Images**
JD Plabon, M Hossain, MA Kabir, D Mistry, MA Hamid
Book: Federated Learning in Health Care Technology. **Publisher:** Springer Nature Singapore
DOI: [10.1007/978-981-96-8353-6_6](https://doi.org/10.1007/978-981-96-8353-6_6)

Interdisciplinary Collaborations

- 2025 P1 **A Multisectoral Study of Mpox Epidemiology, Resistance Surveillance, and Policy Gaps: Toward a One Health Framework (Review Article)**
D Mistry, M Hossain, BS Diba, **JD Plabon**, MA Shams, MT Rahman, ASM Bakibillah
Clinical Reviews in Allergy & Immunology (Q1, Impact Factor: 11.3) - 1 citation
DOI: [10.1007/s12016-025-09121-7](https://doi.org/10.1007/s12016-025-09121-7)

</> Skills

Programming	Python, C++, Java, SQL
Research Areas	Federated Learning, XAI, NLP, LLMs, Deep Learning
Machine Learning	PyTorch, TensorFlow, Hugging Face, Scikit-Learn
Tools	Git, LaTeX, Linux, Docker, VS Code, Jupyter
Web Technologies	Angular, Spring Boot, Django
Languages	Bengali (native), English (professional)

Selected Projects

Gesture Translator

- Developed a real-time ASL-to-English translation system using MediaPipe for hand landmark detection and a custom ML classifier; outputs recognized signs as both text and synthesized audio, enabling accessibility for non-signing users (served as a precursor to the Bangla sign language federated learning research published in J1). 

Presentation Controller

- Built a multimodal hands-free presentation control system combining real-time hand gesture recognition (MediaPipe, OpenCV) with voice command processing (Google Speech API); implements background subtraction and adaptive position prediction for robust performance under varied environments. 

Awards and Achievements

Academic Awards

- **Vice Chancellor's Honor Award** (*X 5: Fall 2023; Spring 2023; Fall 2022; Spring 2022; Spring 2021*)
— An academic award for students who achieved CGPA 3.90+ in a semester.
- **Dean's Honor Award** (*Fall 2021*)
— An academic award for students who achieved CGPA 3.75+ in a semester.

Programming Awards

- **ICPC-Jamilur Reza Choudhury Scholarship Award** (*x3: Fall 2023; Spring 2023; Fall 2022*)
— A prestigious scholarship awarded to top-performing contestants in competitive programming.
- **1st Runner Up**, EEE Tech Fest 2.0 Intra University Programming Contest, 2024
- **2nd Runner Up**, EEE Tech Fest 1.0 Intra University Programming Contest, 2023
- **Champion**, UAP Intra Departmental Hackathon, 2022
- **Champion**, UAP Intra University Programming Contest (Junior Category), 2022

Competitive Programming

- **ICPC Asia Dhaka Regional Contestant**, 2021, 2023, and 2024
- Participated in **10+ National Programming Contests** including NCPC, IUPC (BUET, SUST, IUT, UITS), IUCPC, and more
- Solved **2000+ problems** on online judges including Codeforces, CodeChef, AtCoder, LightOJ, UVA, SPOJ, and CSES
- **Specialist** on Codeforces and **3* Programmer** on CodeChef

Service

Co-Convenor, UAP Programming Contest Club (PCC)

Jul 25 - Present

- Organize and coordinate programming contests, training sessions, workshops and mentor students preparing for ICPC and national-level programming competitions.

President, UAP Programming Contest Club (PCC)

Dec 23 - Jul 24

- Led club operations, organized intra-university programming contests, and mentored members preparing for ICPC and national-level competitions.

President, Robotics Club, CSE Department

Jan 24 - Jul 24

- Directed club activities including robotics workshops, and inter-university competitions.

Volunteer, 45th ICPC World Finals, Dhaka, Bangladesh

Nov 22

- Assisted in operational activities at the most prestigious collegiate programming contest in the world, hosted in Bangladesh for the first time.